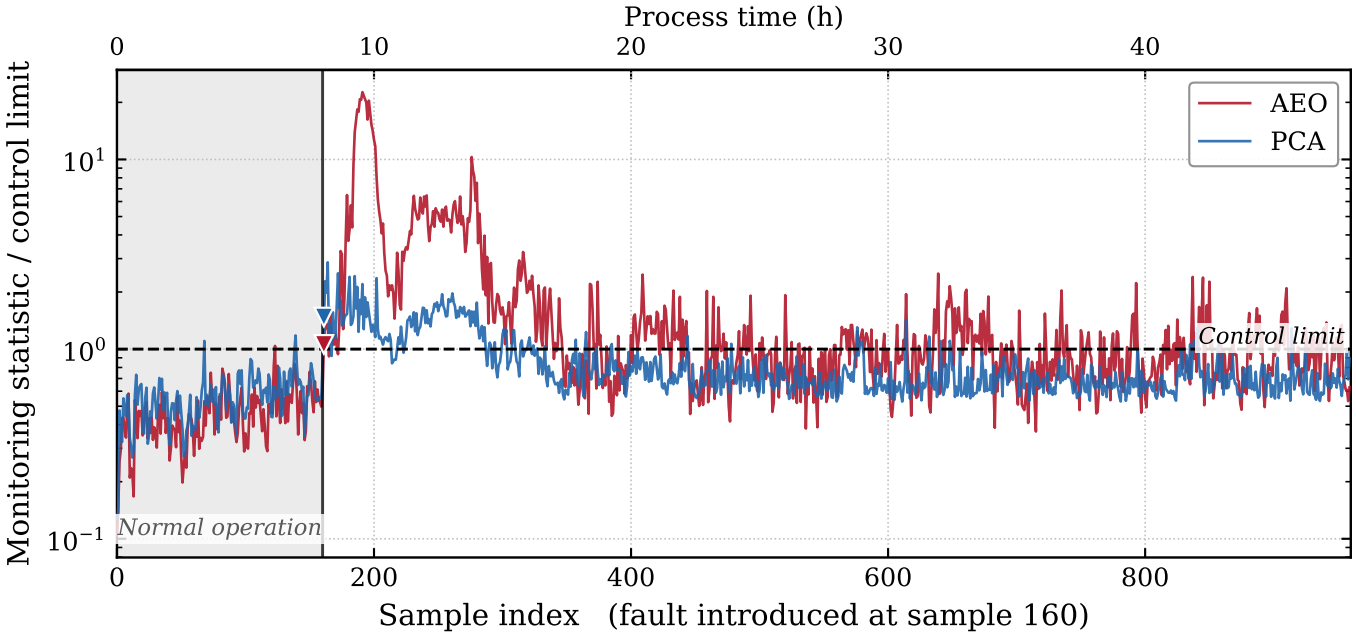
Fault Detection Performance for IDV(5)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	2.4 samples (7 min)	MDR 0.481	detected in 7/7 runs
PCA	delay	0.7 samples (2 min)	MDR 0.728	detected in 7/7 runs

PCA detects 1.7 samples (5 min) earlier; AEO missed-detection rate lower by 34%